

Mohosina Akhter

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Professional Summary

Biomedical engineering and digital health professional with experience in machine learning, healthcare data analysis, and digital health intervention. Recently worked as a Research Assistant at the University of Oulu evaluating predictive performance of machine learning models for patient selection in spinal cord stimulation treatment for chronic pain. Skilled in Python, quantitative analysis, and interdisciplinary collaboration across healthcare and technology domains. Passionate about developing data-driven healthcare solutions and medical technologies with real-world clinical impact.

Technical Skills

Programming & Data Analysis	Biomedical & Digital Health
Python, Machine Learning, Statistical Analysis, Healthcare Data Analysis, Data Visualization	Clinical and Healthcare Data Processing, Predictive Modeling and ML evaluation, Digital Health Systems, Quantitative Research Methods
Tools and Platforms	Professional Skills
Jupyter Notebook, SurveyCTO, Turn.io, Miro, Figma, Git, Microsoft Office Suite	Cross-functional Collaboration, Research Communication, Healthcare Product Design, Development, and Implementation

Education

Master’s in Biomedical Engineering, University of Oulu, Finland. 2026 | GPA: 4.00/5.00
Master’s in Information Technology, Jahangirnagar University, Bangladesh. 2016 | GPA: 3.94/4.00
Bachelor of Science in Computer Science, National University, Bangladesh. 2006 | Second Class (64%)

Master’s Thesis

Evaluating Predictive Performance of Machine Learning Based Patient Selection for Spinal Cord Stimulation in Chronic Pain- Focus on classical machine learning model performance evaluation and robustness analysis

Professional Experience

Research Assistant, Faculty of Information Technology and Electrical Engineering, Software Engineering and Information Systems, SEIS, University of Oulu (July, 2025– May, 2026)

- Evaluating predictive performance of classical machine learning models for patient selection in Spinal Cord Stimulation (SCS) for chronic pain treatment
- Contributing to the design and development of an introductory biomedical engineering course integrating digital health, medical imaging, and hands-on Python applications

Associate Product Manager, Noora Health Bangladesh, Jul 2022 – Oct 2024

- Applied Human-Centered Design methodologies to identify user needs and shape digital health product strategies
- Worked cross-functionally with clinicians, and instructional designers to develop healthcare technology solutions
- Utilized user feedback and data-driven insights to improve intervention performance and usability
- Managed product documentation including experiments, roadmaps, user guides, and training materials

Medical Support Consultant, Noora Health Bangladesh- Dhaka, July 2020 to June 2022

- Managed backend operations for Bangladesh Health Ministry chatbot and digital healthcare communication services
- Maintained healthcare data and supported clinical informatics workflows
- Supported preparation and review of medical content for digital health platforms
- Enhanced digital response accuracy through structured healthcare information management

Program Officer and Information Management Officer, Marie Stopes Bangladesh, Dhaka, December 2015 to June 2020

- Implemented information management solutions and electronic document management systems
- Coordinated healthcare program initiatives and managed MIS data
- Conducted monitoring, evaluation, and impact assessments for healthcare interventions
- Supported healthcare communication and digital transformation activities

Selected Project

LUNA - Sleep Score Optimizer, Course Project | University of Oulu

- Developed a digital health AI system to optimize sleep scores using smart home environmental parameters
- Applied Multiple Linear Regression with Ridge Regularization to model relationships between sleep environment variables and sleep quality outcomes
- Worked with synthetic sleep-environment datasets to analyze temperature and lighting effects on sleep optimization
- Collaborated in a multidisciplinary team environment using GitHub-based project workflows
- Technologies: Python, Scikit-learn, Ridge Regression, GitHub

Certifications and Training

- Object-Oriented Programming (.NET Track), BASIS, Bangladesh (200 hours)
- Leadership Training, Gaurav Singh & Co.lab (4 weeks)
- Survival and Beginners' Finnish Course, University of Oulu

Publications and Research Output

- **Akhter, M., & Isomursu, M. (2026)**
Evaluating Predictive Performance of Machine Learning-Based Patient Selection for Spinal Cord Stimulation in Chronic Pain. In Digital Health and Wireless Solutions: Integrating AI, LLMs and Multimodal Health Data for Next-Generation Decision Support: Proceedings of the Second Nordic Conference on Digital Health and Wireless Solutions (NCDHWS 2026, CCIS 3010, pp. 729–731). Springer Nature. <https://doi.org/10.1007/978-3-032-28819-6>

- **Akhter, M.** (2020).
Changes regarding SRHR knowledge and attitude of secondary institution teachers. Poster presented at the National Conference on Adolescent Health.
- **Akhter, M.,** Hossain, M. S., & Ahmad, F. (2020)
Ensuring safe MR and MRM: Studies conducted in rural areas of Bangladesh. Conference proceedings (ISBN: 978-984-34-8662-2).
- **Akhter, M.,** Hossain, M. S., & Ahmad, F. (2019).
Family planning services and commodities for garment workers in Bangladesh: A situation analysis. International Conference on Global Public Health.

Additional Experience

- Led technical support team of 94 tele-trainers during COVID-19 digital health response
- Managed 21-member healthcare support team at Noora Health
- Conducted onboarding and digital literacy training for hospital staff on mobile healthcare applications

Languages

- Bangla - Native
- English - Professional Proficiency
- Finnish - Beginner

References

Available upon request.